



innovate

achieve

lead

BITS Pilani
Pilani Campus



CHEM F111 : General Chemistry

Semester II: AY 2023-24

Lecture- 01, 10th January 2024, Wednesday

General chemistry - overview of handout

Instructor-in-Charge: Ajay Kumar Sah

Instructors: Ajay Kumar Sah, Anil Kumar, and Avik Kumar Pati

Tutorial Instructors: Anil Kumar, Bharti Khungar, Paritosh Shukla, Shamik Chakraborty, Satyajit Patra, Mrinmoyee Basu, Avik Kumar Pati, Pritam Kumar Jana, and Nitika Grover

Textbooks

T1: P.W. Atkins and Julio de Paula, Elements of Physical Chemistry: 6th Edition, Oxford University Press, Oxford, reprinted in 2015.

T2: T. W. Graham Solomons, Craig B. Fryhle and Scott A Snyder, Organic Chemistry, 12th Edition, John Wiley & Sons, Inc. New York, 2017

Reference books:

R1: J. D. Lee, Concise Inorganic Chemistry, 5th Edition, Blackwell Science, Oxford, 1999.

R2: David Ball, Physical Chemistry, Brooks/Cole Thomson Learning, 2003.

R3: Huheey and Keiter, Inorganic Chemistry: Principles of Structure and Reactivity, 4th Edition,

R4: R. T. Morrison and R. Boyd, 'Organic Chemistry', 6th Edition, PHI, New Delhi, 1992.

Evaluation components



Component	Duration	Weightage % [Marks - 200]	Date and Time (may change)	Remarks
Mid- Sem. Test.	90 min.	30 % [90 M]	12/03/2024	Closed book
Continuous Evaluation [§] (Quiz)	Details later	30 % [90 M]	Continuous	Closed book
Comprehensive Exam.	3 hours	40 % [120 M]	09/05/2024	Partially open book

[§]A total four quiz/assignment (each of 30 marks) will be conducted under continuous evaluation. **Best three** will be considered for final evaluation (i.e., one buffer)

Scope and objectives of the course



The course will provide you:

- A comprehensive survey of various topics in electronic structure of atoms and molecules, spectroscopy, and bonding.
- Understanding of the structure and reactivities of organic and coordination compounds.

Overall objective of the course is to understand fundamental principles of stability of chemical structures and to approach chemical reactions in a systematic and logical way based on electronic structure theory.

1st Part: 16 Lectures

*Quantum theory;
Atomic structure and
chemical bonding;
Spectroscopy: Rotational
& Raman Vibrational*

2nd Part: 9 Lectures

*Coordination compounds
CFT; Distortion of geometries;
Stability of complexes; Electronic
spectroscopy*

3rd Part: 15 Lectures

*NMR spectroscopy; Conformation; Stereochemistry; Types of reactions:
Elimination, Addition, and Substitution; Pericyclic reactions; Aromatic
compounds; Reaction mechanisms*

Quantum Theory: Background

Classical physics

Describes the motion of macroscopic objects, from pendulum, projectiles to parts of machinery, as well as astronomical objects such as spacecrafts, planets, stars, and galaxies.

Classical physics: consequences

1. Predict a precise trajectory for particles with precisely specified locations and momenta at each instant
 2. Any kind of motion can be excited to any arbitrary value of the energy
 3. Waves and particles are distinct concepts
- *Unaltered for three centuries till the end of 19th Century*

- ☐ Black body radiation
- ☐ Photoelectric effect
- ☐ Electron diffraction
- ☐ Line spectra of atoms

The failures of classical mechanics in explaining certain phenomena such as the listed above paved the way for quantum mechanics

Black body



An opaque object that is a perfect absorber and a perfect emitter

At room temperature, such an object would appear to be perfectly black

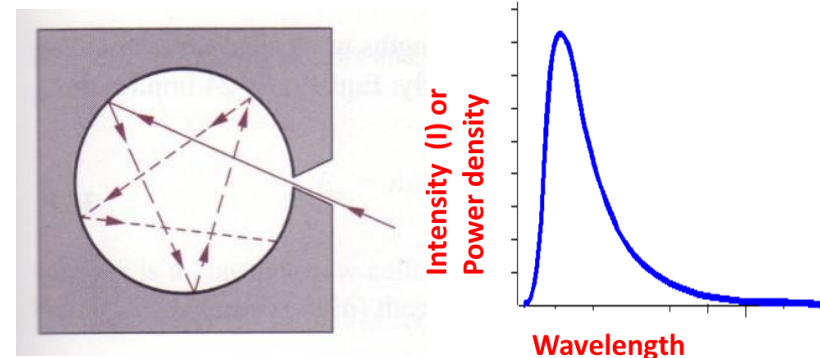
If heated to a high temperature, it will begin to glow and emit a continuous spectrum composed of all wavelengths of light (thermal radiation)

The distribution of absorbed or emitted radiation depends on the absolute temperature, not on the body material

Not all the wavelengths of light are emitted with equal intensity

At constant temperature, intensity increases as λ increases, attains a maximum value and then decreases

Actual black bodies don't exist in nature



Stefan-Boltzman observations: 1884



The **Stefan–Boltzmann law** describes the power radiated from a blackbody in terms of its temperature.

Stefan-Boltzman law

$$M = \sigma T^4$$

Stefan-Boltzman constant

$$\sigma = 5.6697 \times 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$$

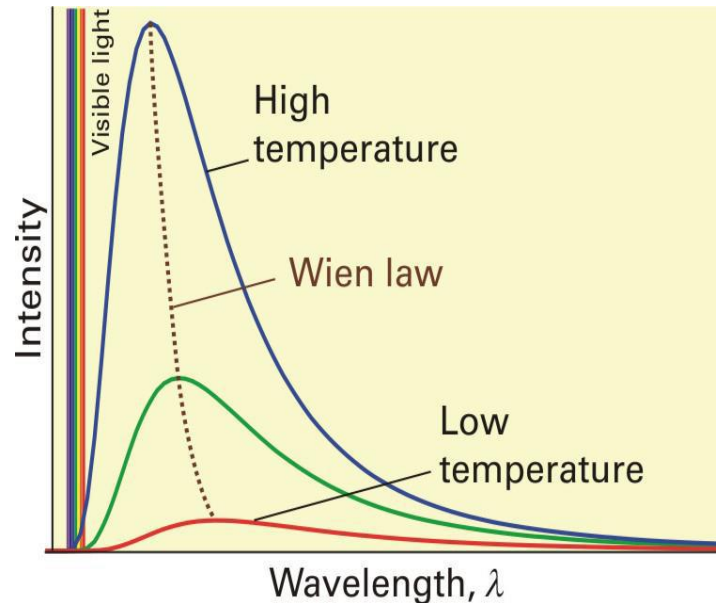
- **Emittance M:** Total power emitted per unit area
- M is proportional to 4th power of absolute temperature
- M is expressed in W/m² unit

The Sun at 5800 K and a hot campfire at perhaps 800 K give off radiation at a rate proportional to the 4th power of the temperature



Wien's displacement law

$$\lambda_{max} T = b; \quad b = 2.9 \text{ mmK} = 0.29 \text{ cmK}$$



- With increase in T , the λ_{max} shifts towards shorter wavelength
- The wavelength of maximum emission from a blackbody is inversely proportional to its temperature.

I. Rayleigh-Jeans explanation (1900-1905)

The Rayleigh-Jeans Radiation Law was useful but not completely a successful attempt at establishing the functional form of the spectra of thermal radiation through classical arguments.

This attempt was based on certain assumptions (believed to be true at that time).

Assumptions:

- ✓ Blackbody cavity is made up of charged particles which behaves as tiny oscillators by thermal accelerations and emit radiations.
- ✓ Energy emitted by atomic oscillators could have any continuous value.

Rayleigh-Jeans law



Energy density [$dE = \rho(\lambda)d\lambda$] is the energy per unit volume associated with radiation of wavelength from λ to $\lambda + d\lambda$:

$$dE = \rho d\lambda = \frac{8\pi}{\lambda^4} kT d\lambda$$

k is Boltzmann's constant

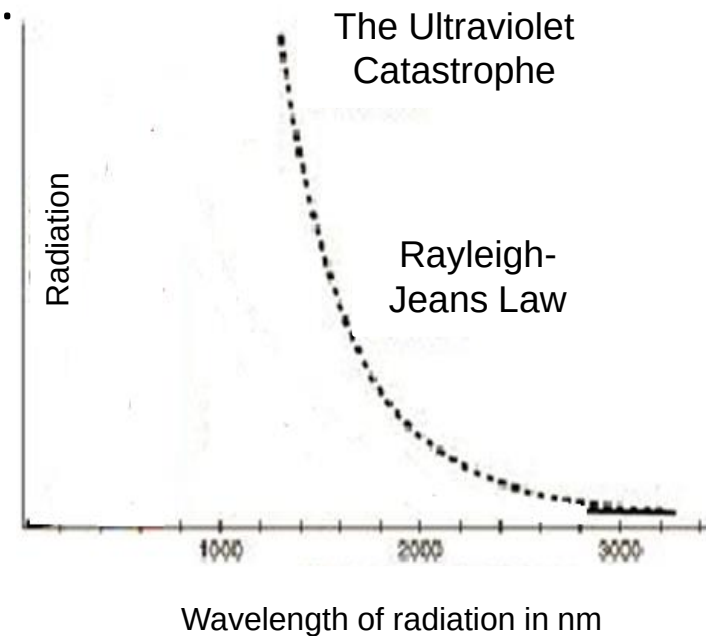
Infinite energy density at short wavelengths as energy density raises without bound as λ decreases!

*The law predicts that oscillators of short wavelength (UV) are excited even at room temperature.: **This cannot be true***

*This paradox: **ULTRAVIOLET CATASTROPHE!***

Quite successful at long wavelengths

Unsuccessful attempt to explain the complete blackbody radiation spectrum

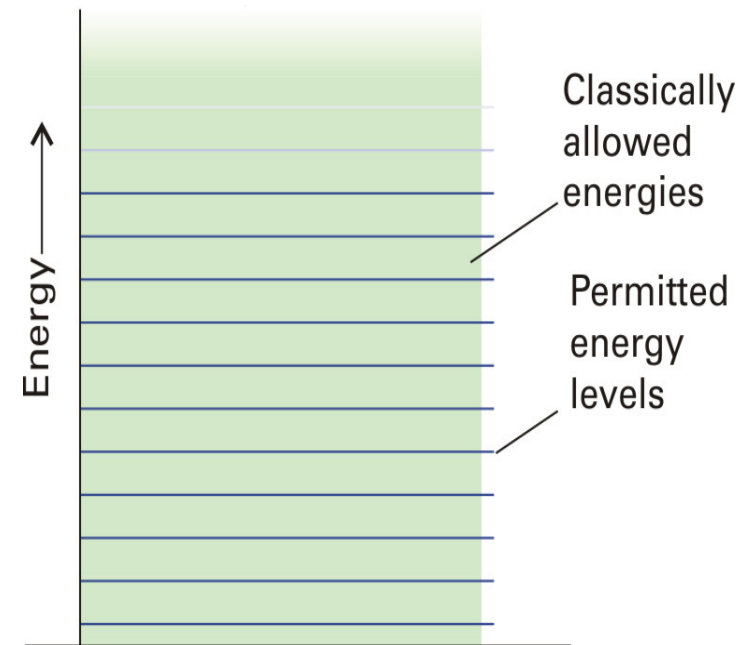


II. Planck's Explanation (1900-1905)

Attempts to describe the emission spectrum from a black body at a given temperature through quantization hypothesis.

Crucial Assumption:

An oscillator of frequency ν **cannot be excited to any arbitrary energy**, but to only to **integral multiples of a fundamental unit or quantum of energy $h\nu$** ; $h = 6.626 \times 10^{-34}$ Js, the Planck's constant, i.e. energy of an oscillator is quantized

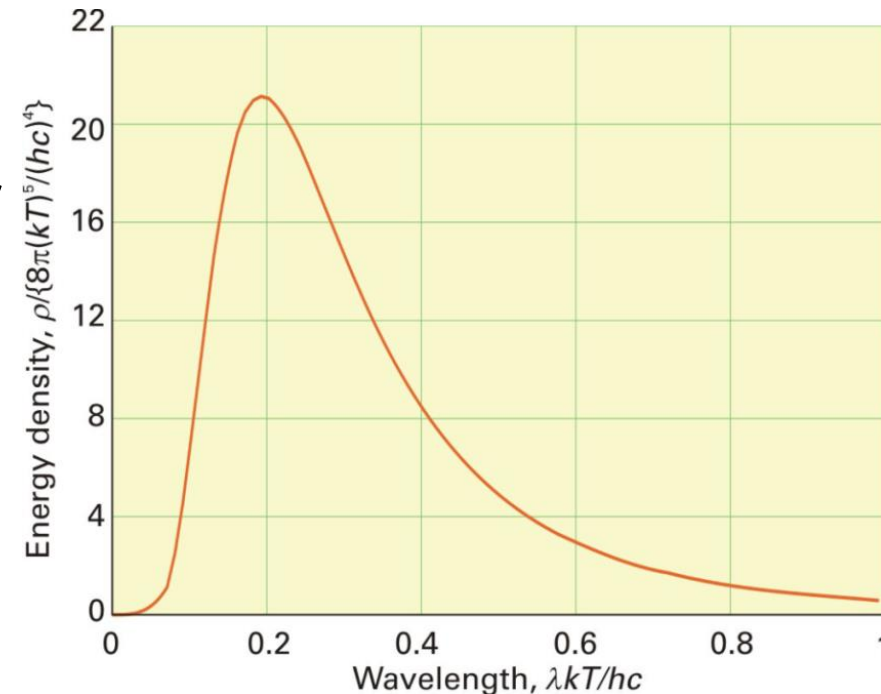


$$dE = \rho(\lambda)d\lambda = \frac{8\pi}{\lambda^4} \frac{hc/\lambda}{e^{hc/\lambda kT} - 1} d\lambda$$

c is speed of light; k is Boltzmann's constant and h is Planck's constant.

Planck's proposed formula describes the curve of blackbody radiation exactly for all wavelengths.

Planck expression reproduces the experimental distribution with $h = 6.626 \times 10^{-34}$ Js.



Planck law: success story

- ✓ Planck's hypothesis: An oscillator cannot be excited unless it receives an energy of at least $h\nu$ (as this is the minimum amount of energy an oscillator of frequency ν may possess above zero).
- ✓ For high frequency oscillators (large ν , low λ), the amount of energy $h\nu$ is too large to be supplied by the thermal motion of the atoms in the walls, and so they are not excited.

$$dE = \frac{8\pi}{\lambda^5} \frac{hc}{e^{hc/\lambda kT} - 1} d\lambda$$

At small λ , $e^{hc/\lambda kT} \rightarrow \infty$ faster than λ^5
(Exponential is large)
Energy density $\rightarrow 0$ as $\lambda \rightarrow 0$

**UV Catastrophe
avoided**

Puzzling! The explanation of blackbody radiation is successful only if we assume that emitted energy from the oscillators is discrete. Do you have a simple clarification on how the discreteness of energy does the job??



$$dE = \rho d\lambda = \left(\begin{array}{c} \text{number of modes} \\ \text{per unit volume with} \\ \text{wavelengths between} \\ \lambda \text{ and } \lambda + d\lambda \end{array} \right) \times \left(\begin{array}{c} \text{average energy} \\ \text{of a mode} \end{array} \right)$$

$$= \frac{8\pi}{\lambda^4} d\lambda \times \left(\begin{array}{c} \text{average energy of} \\ \text{a mode} \end{array} \right)$$

Two arrows point from the 'average energy of a mode' term to the following expressions:

- Rayleigh-Jeans: kT
- Planck: $\frac{hc/\lambda}{e^{hc/\lambda kT} - 1}$



The magic of Planck's formula

$$M = \frac{1}{4} Ec = \frac{ac}{4} T^4 = \sigma T^4$$

$$\sigma = 5.6697 \times 10^{-8} \text{ J s}^{-1} \text{ m}^{-2} \text{ K}^{-4}$$

$$E = aT^4; a = 7.566 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4}$$

Stefan-Boltzmann law

$$\lambda_{\max} T = \frac{hc}{5k} \text{ (constant)}$$

Wien's law

$\frac{d\rho}{d\lambda} = 0$ for calculating $\lambda_{\max}$ (at low λ)

$$E = \int_0^{\infty} \rho(\lambda, T) d\lambda$$

$$dE = \rho d\lambda = \frac{8\pi}{\lambda^4} \frac{hc/\lambda}{e^{hc/\lambda kT} - 1} d\lambda$$

$$e^{hc/\lambda kT} = 1 + \frac{hc}{\lambda kT} + \frac{1}{2} \left(\frac{hc}{\lambda kT} \right)^2 + \dots$$

Dropping

For long wavelengths,

$$\frac{hc}{\lambda kT} \ll 1$$

Rayleigh-Jeans formula